

Machine Learning for Biology and Health

CSCI 1851
Spring 2026

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Tuesday



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Today's goal – Continue to learn about kernels and support vector machine (SVM)

- (1) Linear SVMs with soft margins
- (2) Dual form of SVM
- (3) Revisiting Kernels
- (4) Kernelized SVMs
- (5) Class activity: Kernelized SVM

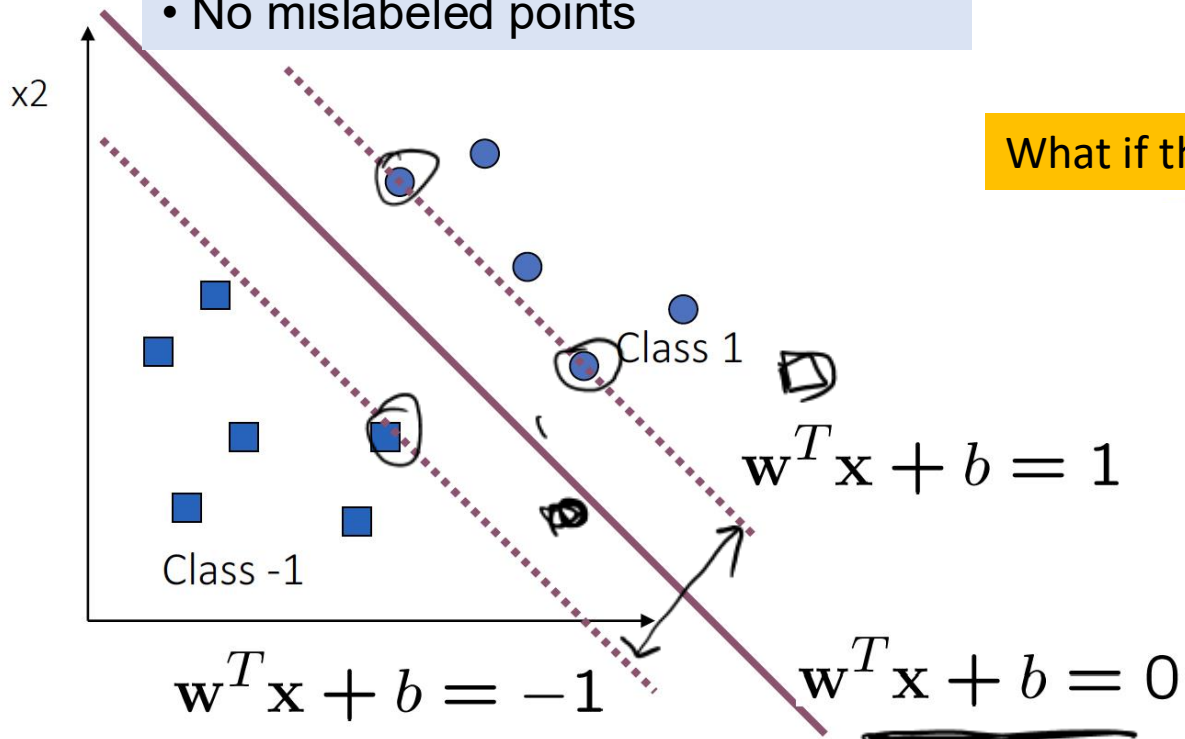
Linear SVM with soft margins

Review: SVM optimization

1. Correctly classifies all points
2. Maximizes the margin (or equivalently minimizes $w^T w$)

Works well when,

- Data is perfectly linearly separable
- No noise
- No mislabeled points



$$\begin{aligned} \min_{w,b} \quad & \frac{1}{2} \|w\|^2 \\ \text{subject to} \quad & y_i (w^T x_i + b) \geq 1, \quad \forall i \end{aligned}$$

Why square the L2-norm?

- Convex
- Smooth
- Quadratic

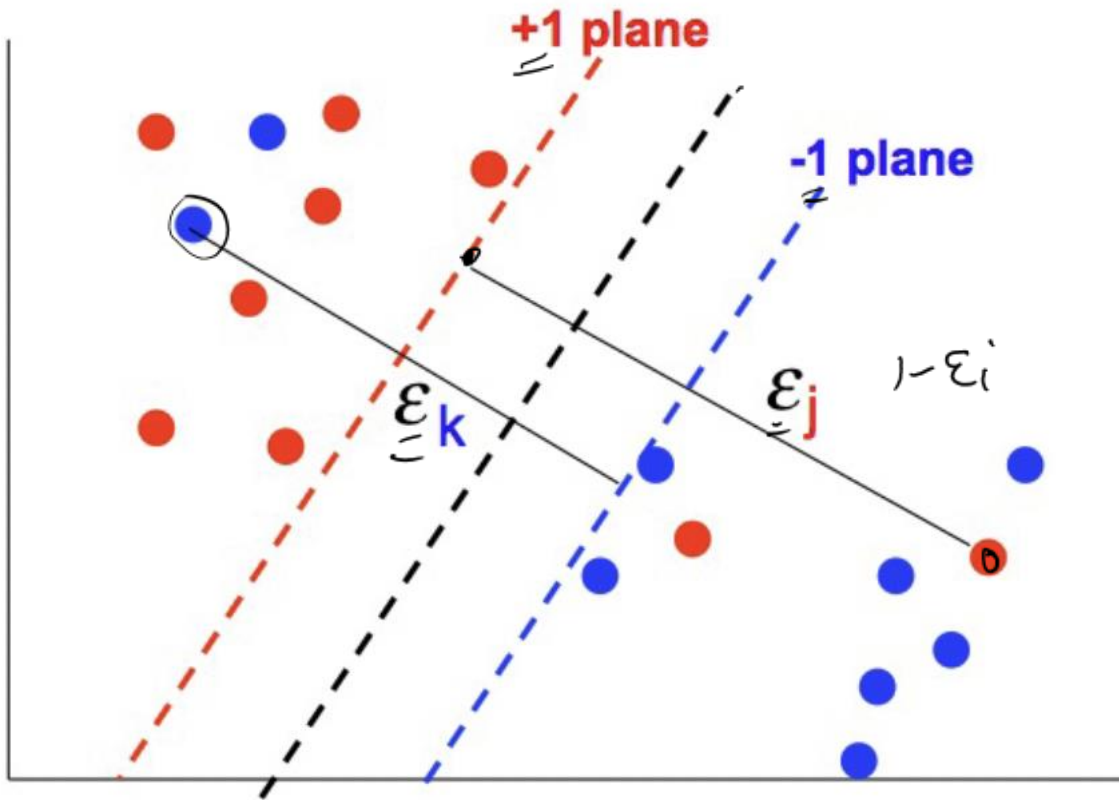
What if the data is noisy??

Quadratic programming i.e.,

- Quadratic objective
- Linear constraints

Going from hard margin to soft margin

- Instead of minimizing the number of misclassified points we can **minimize the distance between these points and their correct plane**



The new optimization problem is:

$$\min_w \left\{ \frac{w^T w}{2} + C \sum_{i=1}^n \varepsilon_i \right\}$$

Slack Variable

subject to the following inequality constraints:

$$\left\{ \begin{array}{l} \text{For all } x_i \text{ in class } +1 \\ w^T x_i + b \geq 1 - \varepsilon_i \\ \text{For all } x_i \text{ in class } -1 \\ w^T x_i + b \leq -1 + \varepsilon_i \end{array} \right.$$

For all i

$$\varepsilon_i \geq 0$$

New SVM Optimization

Hard Margin

$$\begin{aligned} \min_{w,b} \quad & \frac{1}{2} \|w\|^2 \\ \text{subject to} \quad & y_i(w^T x_i + b) \geq 1, \quad \forall i \end{aligned}$$

Soft Margin

$$\begin{aligned} \min_{w,b,\xi} \quad & \frac{1}{2} \|w\|^2 + \underline{C} \sum_i \xi_i \\ & \text{subject to} \\ & y_i(w^T x_i + b) \geq 1 - \xi_i \\ & \xi_i \geq 0 \end{aligned}$$

(Handwritten notes: an arrow points from ξ_i to the word "error" below it, and a wavy line is under the ξ_i in the objective function)

C is a regularization parameter

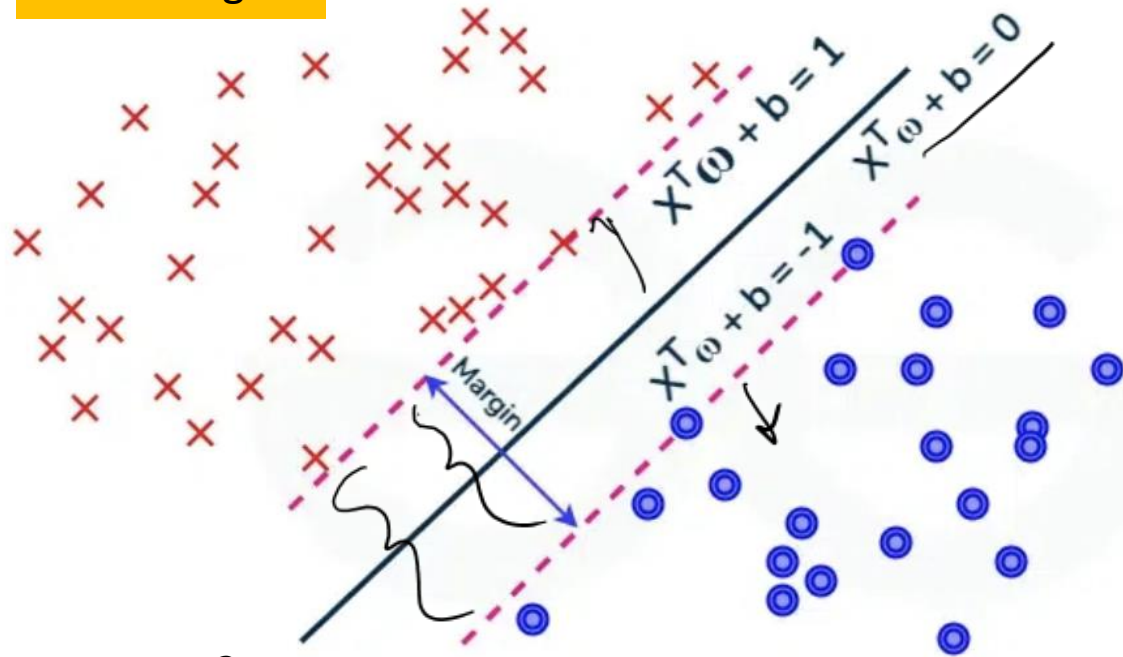
Small C $\rightarrow$ wide margin, more tolerance (high bias)

Large C $\rightarrow$ narrow margin, less tolerance (low bias)

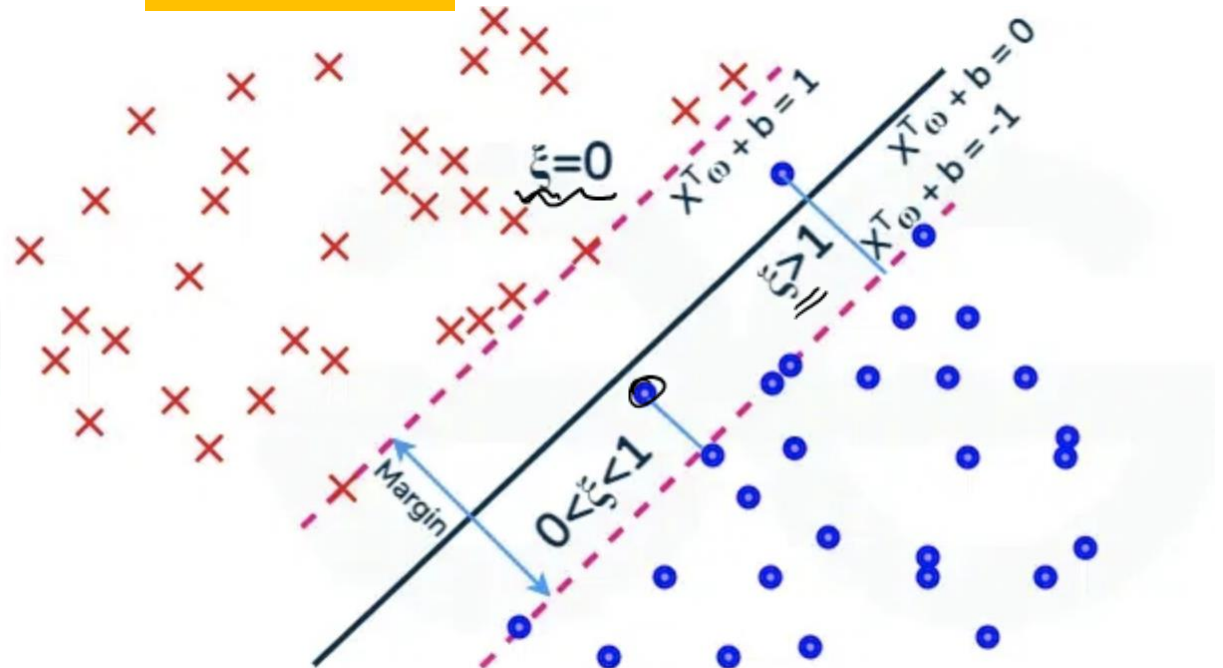
Linear SVM with soft margins

What if the data is not linearly separable?

Hard Margin



Soft Margin



Any questions?



Works well when data is linearly separable (not perfectly so)

Can handle noise and mislabeled points

Dual form of SVM

Duality principle

In mathematical optimization theory, **duality** or the **duality principle** is the principle that optimization problems may be viewed from either of two perspectives, the **primal problem** or the **dual problem**.

If the primal is a minimization problem, then the dual is a maximization problem (and vice versa).

Why convert to dual form?

- Constraints become relaxed
- The data appears only in dot products
- That will allow kernels



$$\min_{w,b,\xi} \frac{1}{2} \|w\|^2 + C \sum_i \xi_i$$

$$y_i(w^T x_i + b) \geq 1 - \xi_i \quad ?$$

$$\xi_i \geq 0 \quad ?$$

*primal
problem*

Lagrangians as a "penalty trick" (simple example)

$$\text{minimize } x^2 \text{ subject to } x \geq 1$$

$$\begin{array}{l} x \geq 1 \\ (x-1) \geq 0 \\ \alpha (x-1) \geq 0 \\ \alpha \geq 0 \end{array} \quad \alpha = 0$$

$$L(x, \alpha) = x^2 - \alpha(x-1)$$

Lagrange multiplier

$$\text{Here, } \alpha \geq 0$$

Lagrange multipliers enforce constraints by making their violation infinitely costly.

The dual problem re-expresses the same optimization from the perspective of those costs rather than the original variables.

$$\min_x \max_{\alpha} L(x, \alpha) = \min_x \text{ subject to constraints}$$

$$\max_{\alpha} L(\alpha) \rightarrow \text{dual form}$$

For convex problems, both perspectives reach the exact same answer.

$$\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum_i \xi_i$$

$$y_i(w^T x_i + b) \geq 1 - \xi_i$$

$$\xi_i \geq 0$$

$$y_i(w^T x_i + b) - 1 + \xi_i \geq 0$$

$$\alpha_i [y_i(w^T x_i + b) - 1 + \xi_i] \geq 0$$

$$\xi_i \geq 0$$

$$\mu_i \xi_i \geq 0$$

max L
 α, μ

$$\min_{w, b, \xi} \max_{\alpha, \mu} \mathcal{L} = \frac{1}{2} \|w\|^2 + C \sum_i \xi_i - \sum_i \alpha_i [y_i(w^T x_i + b) - 1 + \xi_i] - \sum_i \mu_i \xi_i$$

Lagrange multipliers

$\alpha_i \geq 0$ for margin constraints

$\mu_i \geq 0$ for slack constraints

$$L = \frac{\partial L}{\partial w} \quad w - \sum_i \alpha_i y_i x_i = 0 \rightarrow \textcircled{1}$$

w, b, ε

$w = \sum_i \alpha_i y_i x_i$ } weighted sum of training points.

$$\frac{\partial L}{\partial b} \quad \sum_i \alpha_i y_i = 0 \quad \text{ } \} \text{ constraint } \quad \textcircled{2}$$

$$\frac{\partial L}{\partial \varepsilon} \quad c - \alpha_i - \mu_i = 0 \quad \mu_i \geq 0$$

$$c \geq \alpha_i = \mu_i$$

$$c \geq \alpha_i \geq 0$$

$$\alpha_i \leq c$$

$$0 \leq \alpha_i \leq c \quad \text{ } \} \text{ constraint. } \quad \textcircled{3}$$

New dual SVM formulation

What is interesting here?

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \underbrace{x_i^T x_j}_{\substack{\phi(x_i) \cdot \phi(x_j) \\ K(x_i, x_j)}} =$$

subject to

$$0 \leq \alpha_i \leq C$$

$$\sum_i \alpha_i y_i = 0$$

$$\alpha_i > 0$$

support vectors

$$\alpha_i = 0$$

data points not my support vectors.

$$\alpha_i [y_i (w^T x_i + b) - 1 + \epsilon_i] \geq 0$$

$\downarrow$
 $= 0$

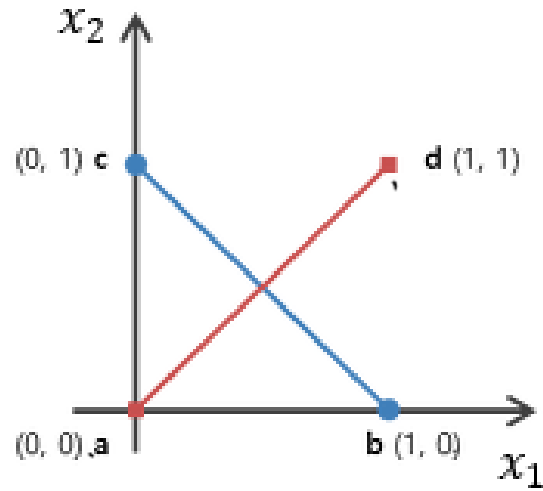
Any questions?

???



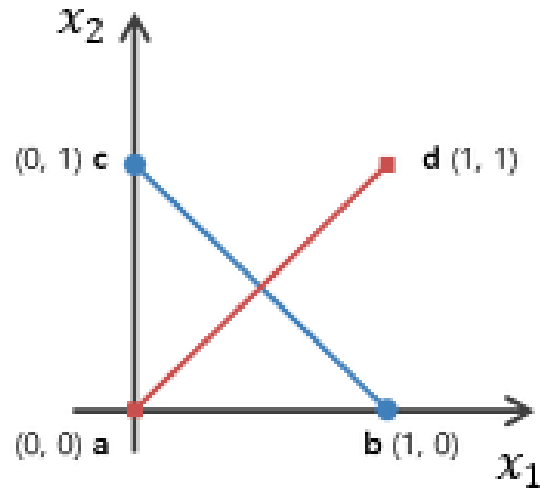
Revisiting Kernels

Projecting data in higher dimension versus the kernel trick



Projecting data in higher dimension versus the kernel trick

Polynomial feature expansion



$[x_1, x_2]$
 $[z_1, z_2]$

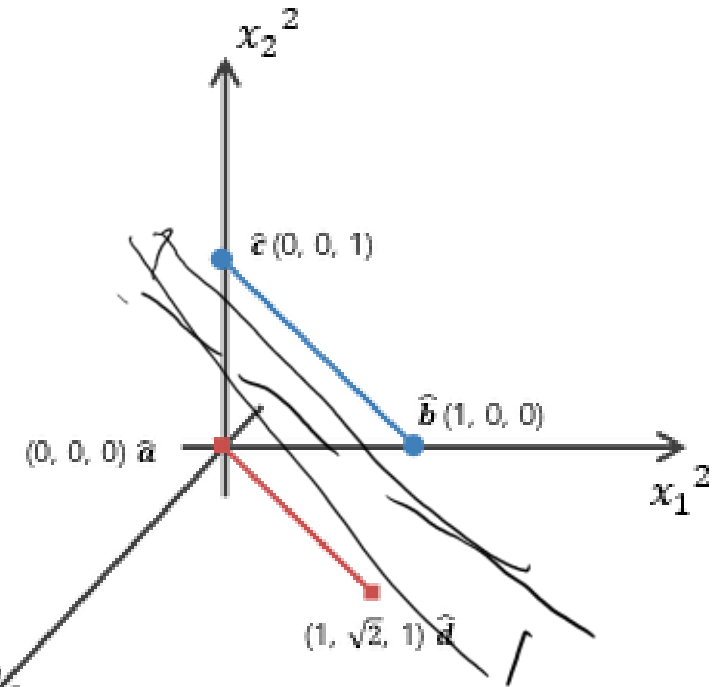
$$\varphi(x) = \begin{pmatrix} x_1^2 \\ \sqrt{2}x_1x_2 \\ x_2^2 \end{pmatrix}$$

2 points x, z

polynomial kernel
 $[=(xz)^d]_{d=2}$

$$K = x_1^2 z_1^2 + 2x_1 x_2 z_1 z_2 + x_2^2 z_2^2$$

$$X = \left((x_1^2, \sqrt{2}x_1x_2, x_2^2) \left(z_1^2, \sqrt{2}z_1z_2, z_2^2 \right) \right)$$



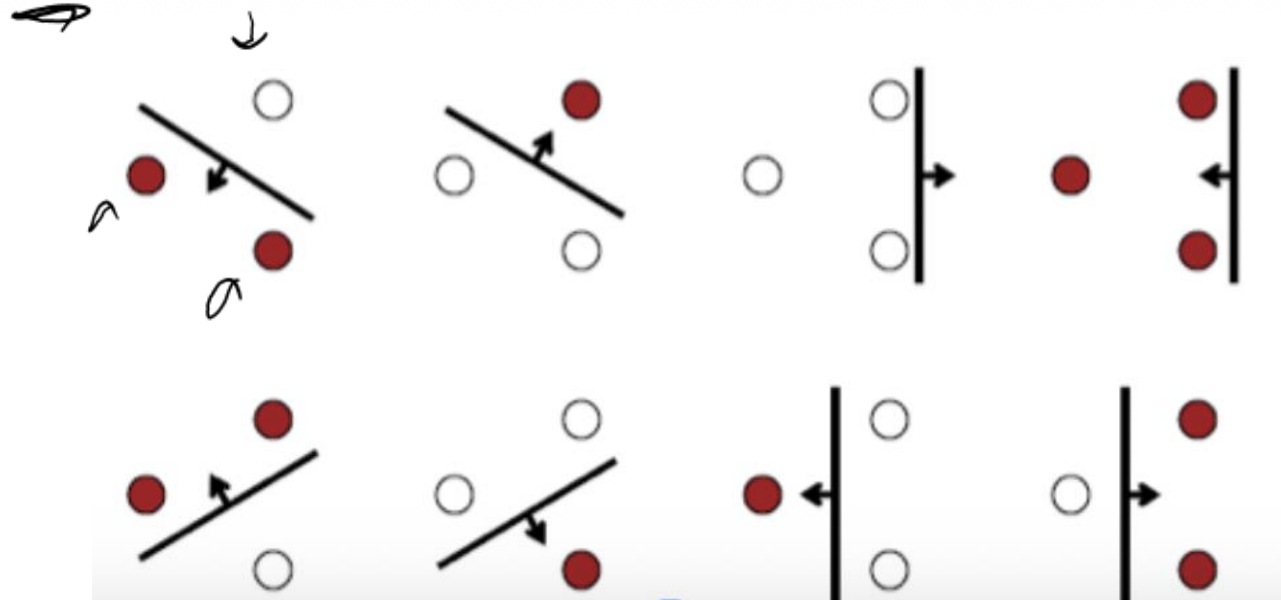
A little bit of theory

Vapnik-Chervonenkis (VC) dimension

If data is mapped into sufficiently high dimension, then samples will in general be linearly separable;



- VC dimension of the set of oriented lines in $\mathbb{R}^2$ is 3
 - It can be shown that the VC dimension of the family of oriented separating hyperplanes in $\mathbb{R}^N$ is at least $N+1$



Different examples of kernels

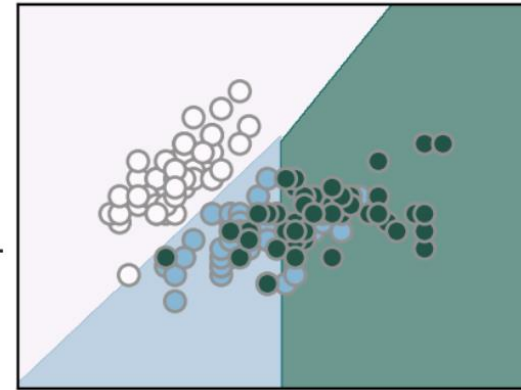
$$K(x_i, x_j) = \underline{\underline{x_i^T x_j}}$$

$$K(x_i, x_j) = (x_i^T x_j + c)^d$$

$$\underline{\underline{K(x_i, x_j)}} = \exp(-\gamma \underline{\underline{\|x_i - x_j\|^2}})$$

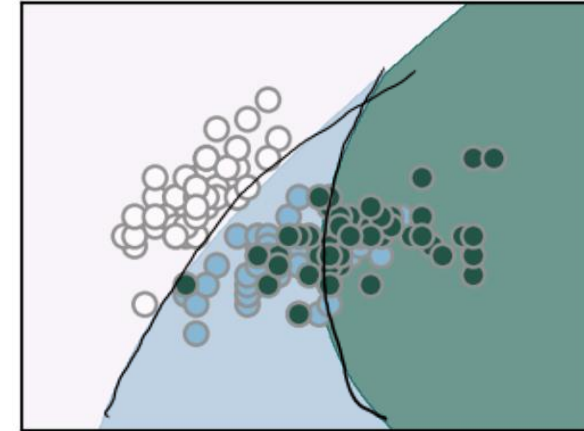
RBF similarity is basically a smooth version of "are these two points neighbors?". SVM builds its decision boundary by asking that question about every training point.

Linear kernel



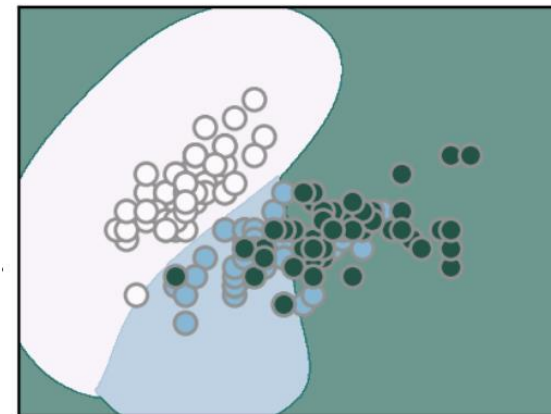
Sepal length

Polynomial kernel



Radial Basis Function.

RBF kernel



Any questions?



Kernelized SVMs

SVM with kernel trick

- Change all inner products to kernel functions

- For training,

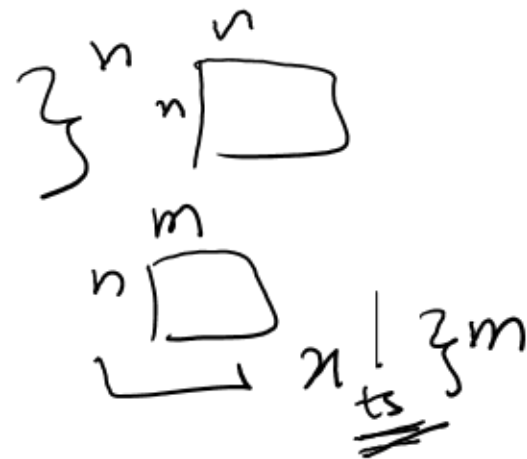
Original
Linear

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j$$

$$\sum_i \alpha_i y_i = 0$$

$$C > \alpha_i \geq 0, \forall i \in \text{train}$$

Training

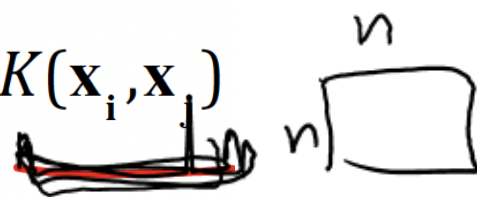


With kernel
function -
nonlinear

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j K(\mathbf{x}_i, \mathbf{x}_j)$$

$$\sum_i \alpha_i y_i = 0$$

$$C > \alpha_i \geq 0, \forall i \in \text{train}$$



SVM with kernel trick

- For testing, the new data $\mathbf{x}_{ts}$

Original
Linear

$$\widehat{y}_{ts} = \text{sign} \left(\sum_{i \in \text{supportVectors}} \alpha_i y_i \mathbf{x}_i^T \mathbf{x}_{ts} + b \right)$$

Handwritten notes: x_i^T above the sum, $\omega^T x_{ts}$ above the term $\mathbf{x}_i^T \mathbf{x}_{ts}$, and $\mathbf{x}_i$ and $\mathbf{x}_{ts}$ are underlined.

With kernel
function -
nonlinear

$$\widehat{y}_{ts} = \text{sign} \left(\sum_{i \in \text{supportVectors}} \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}_{ts}) + b \right)$$

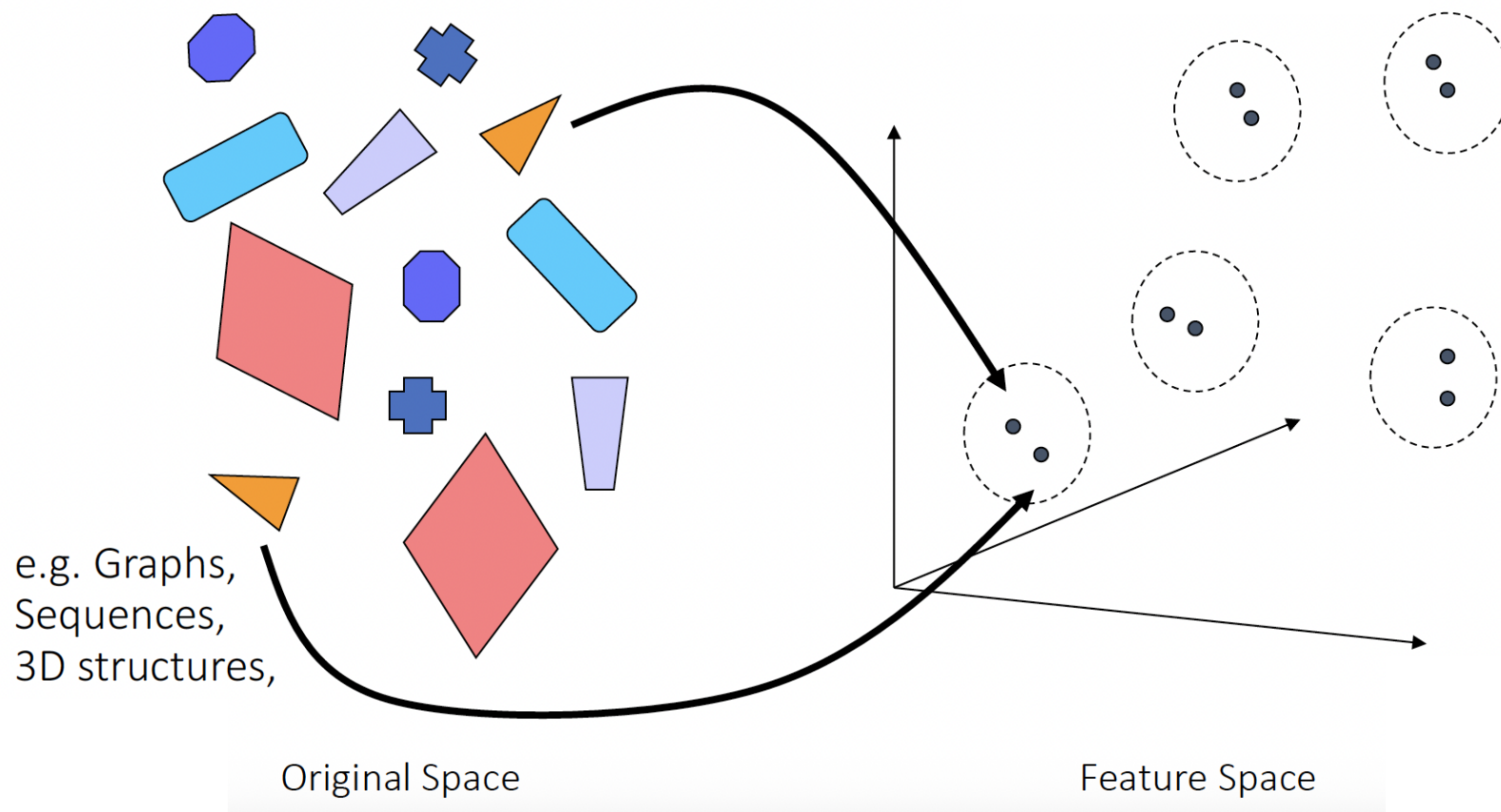
Handwritten notes: x_i^T above the sum, $K(\mathbf{x}_i, \mathbf{x}_{ts})$ is circled, and $\mathbf{x}_i$ and $\mathbf{x}_{ts}$ are underlined.

Kernel trick has helped Non-traditional data like strings and trees able to be used as input to SVM, instead of feature vectors

Any questions?



Vector vs. Relational data



Class activity

Kernelized Support Vector Machine

Recap – Learn about kernels and support vector machine (SVM)

(1) Linear SVMs with soft margins

(2) Dual form of SVM

(3) Revisiting Kernels

(4) Kernelized SVMs

(5) Class activity: Kernelized SVM

Homework Reading:
Sections 7.7, 11.5, 11.6
[in CML Book](#)

Wrap up



What was the clearest point today?

What was the muddiest point today?

